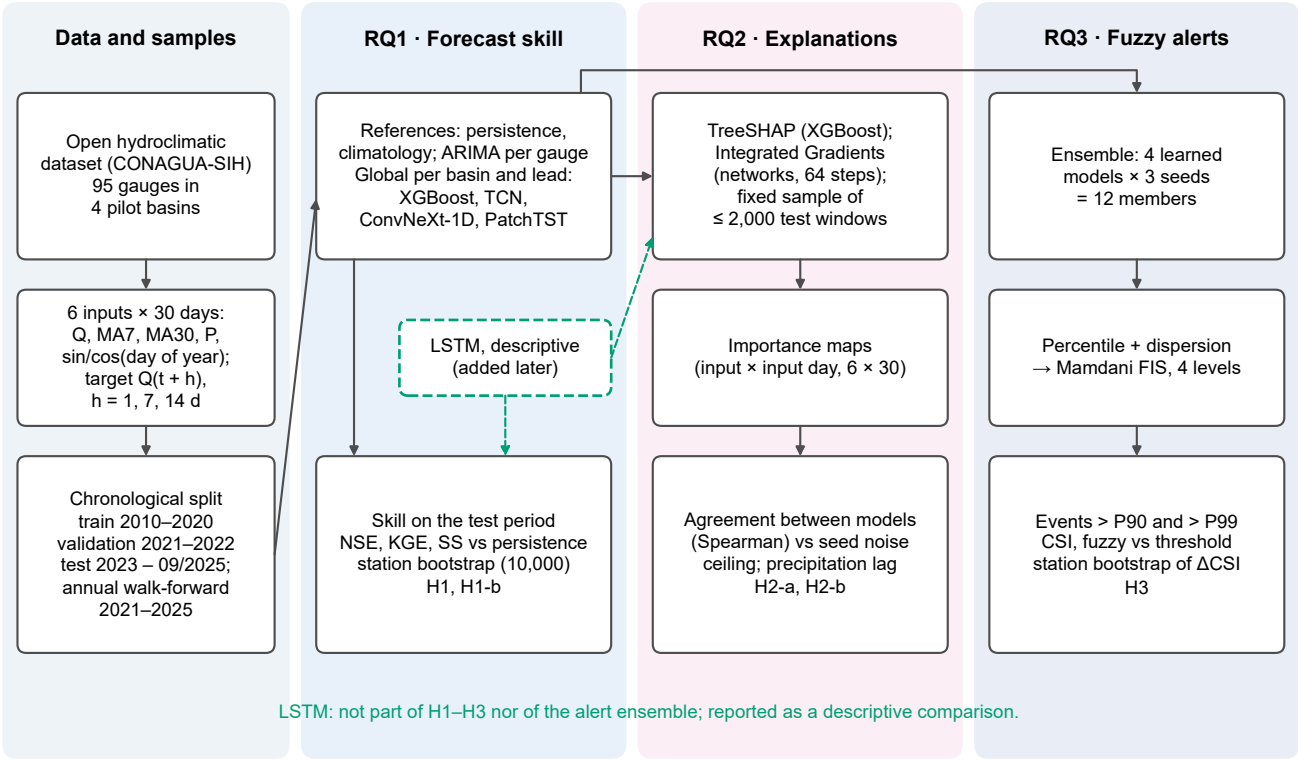

Data and samples
RQ1 · Forecast skill
RQ2 · Explanations
RQ3 · Fuzzy alerts
Open hydroclimatic dataset (CONAGUA-SIH) 95 gauges in 4 pilot basins
6 inputs × 30 days: Q, MA7, MA30, P, sin/cos(day of year); target Q(t + h), h = 1, 7, 14 d
Chronological split train 2010–2020 validation 2021–2022 test 2023 – 09/2025; annual walk-forward 2021–2025
References: persistence, climatology; ARIMA per gauge Global per basin and lead: XGBoost, TCN, ConvNeXt-1D, PatchTST
LSTM, descriptive (added later)
Skill on the test period NSE, KGE, SS vs persistence station bootstrap (10,000) H1, H1-b
TreeSHAP (XGBoost); Integrated Gradients (networks, 64 steps); fixed sample of ≤ 2,000 test windows
Importance maps (input × input day, 6 × 30)
Agreement between models (Spearman) vs seed noise ceiling; precipitation lag H2-a, H2-b
Ensemble: 4 learned models × 3 seeds = 12 members
Percentile + dispersion → Mamdani FIS, 4 levels
Events > P90 and > P99 CSI, fuzzy vs threshold station bootstrap of ΔCSI H3
LSTM: not part of H1–H3 nor of the alert ensemble; reported as a descriptive comparison.